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- MAJ Shrestha

ACI Research & Independent Study
AI and Targeting

Year One

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Exploring AI and Targeting

Problem Statement

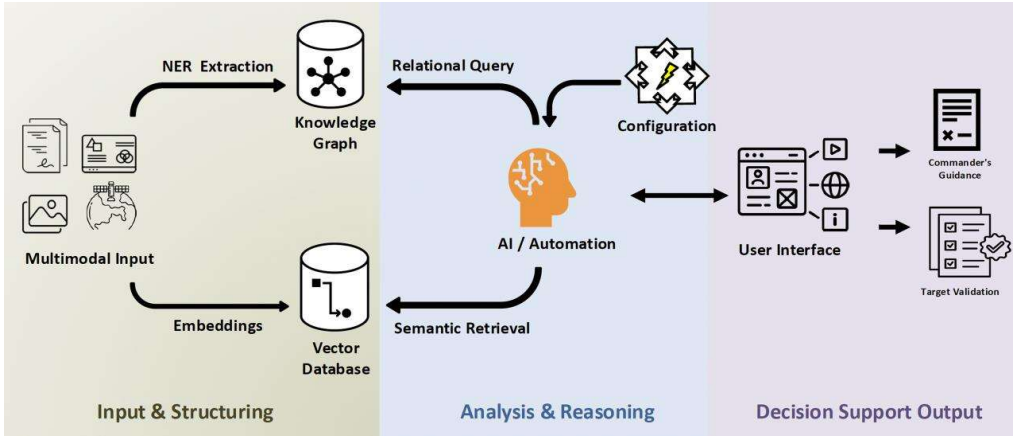
- **State of the Art:** AI is increasingly used in military decision workflows, especially in time constrained environments like targeting. The goal is not just accurate outputs. Sustained trust requires structured human AI interaction, traceability, and the ability to review and audit decisions over time.
- **Gaps:** Clear, operational guidance remains limited, and organizations still struggle to translate high-level expectations into practical requirements for oversight, audit, and reviewable evidence in real operations.
- **Possible Solutions:** Develop a workflow embedded AI prototype aligned with doctrinal decision stages. Instrument interactions at key checkpoints to enable structured evaluation of human AI teaming, traceability, and oversight in realistic settings.

Project Deliverables

- **Stakeholder(s): DAMO-SO, ARCYBER, DA G2**
- **Deliverable Products:**
  - Publications and technical reports based on findings. (Talk at MORS AI Workshop 2026 and paper in final review)
  - Workflow embedded prototype for AI assisted targeting decision support
- **Expected Outcomes:**
  - Empirical insight based on user studies into human AI interaction at decision checkpoints
  - Identification of feasibility limits, workflow friction points, and oversight needs.

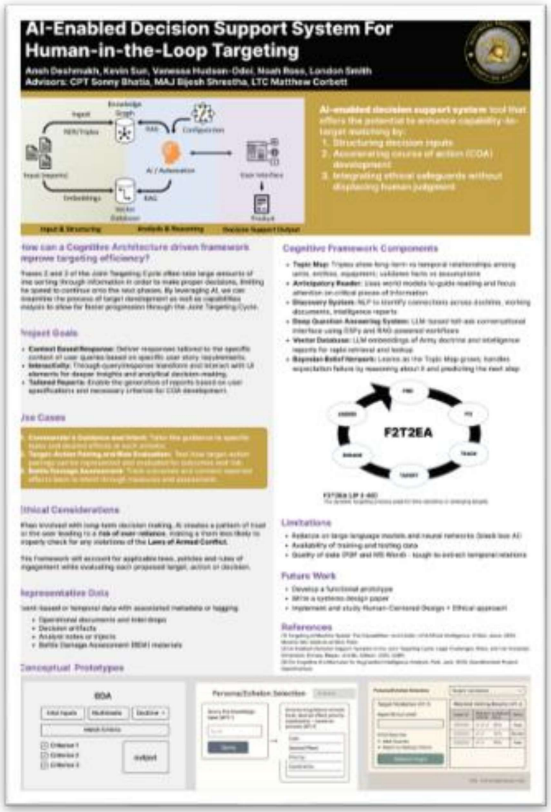
Project Description

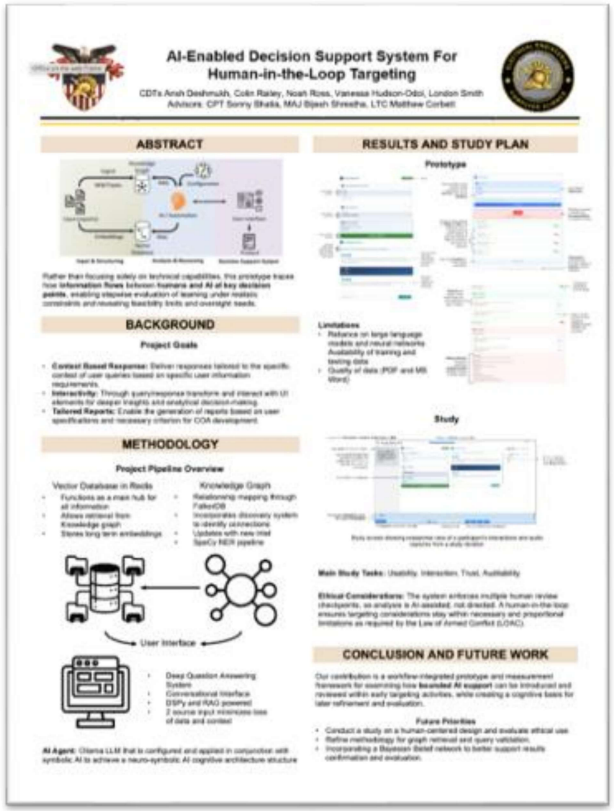
- **Goal**: Explore how AI can support targeting workflow related tasks without displacing human judgement, by embedding AI within structured checkpoints that enable review, verification, and traceability.
- **Methods**:
  - Design a human AI workflow aligned with targeting stages (D3A)
  - Develop a prototype that can be refined and iterated through user studies and design studies on usability, trust, and auditability.
  - Integrate technical solutions that enable authoritative interrogation of the data corpus.
- **Implications**: Provides a workflow grounded approach to evaluating AI in high stakes environments, shifting from isolated model performance to accountable, human centered decision support.



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Outputs







Exploring AI for Decision Support in Targeting

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***Abstract*— The rapid advancement of artificial intelligence presents new opportunities to support military decision-making. Operational leaders face increasingly complex environments where large volumes of information must be interpreted quickly and accurately. While AI tools promise to assist with processing and analysis, these workflows remain time-sensitive and dependent on human judgment and accountability. Emerging approaches to human–AI collaboration emphasize structured interaction, transparency, and staged evaluation, suggesting that AI can support analysis while leaders retain decision authority and contextual judgment. This work presents an early-stage exploration of AI support for targeting decision cycles through a prototype that interprets structured orders to enable target validation. Rather than focusing solely on technical capabilities, this prototype traces how information flows between humans and AI at key decision points, enabling stepwise evaluation of teaming under realistic constraints and revealing feasibility limits and oversight needs. By emphasizing workflow integration rather than isolated demonstrations, this research treats AI-enabled targeting as a structured research question with a concrete use case and defined checkpoints to study feasibility, limits, and oversight.**

I. Introduction

The integration of artificial intelligence (AI) into military decision-making has accelerated in recent years, particularly in domains characterized by high information density and compressed timelines. Targeting is one such domain that requires high-risk decisions about military effects and weapons in extremely time-constrained environments. Joint Publication (JP) 3-60, *Joint Targeting Cycle* (JTC), codifies targeting as a structured, cyclical function embedded within joint operations, while Field Manual (FM) 3-60, *Army Targeting*, translates that construct into echelon-specific practice [1], [2]. Across these publications, targeting is defined not merely as objective identification but also as the deliberate integration of intelligence, operations, fires, and assessment to achieve commander-directed effects under legal and operational constraints. The U.S. Army uses the Decide, Detect, Deliver, Assess (*D3A*) methodology as its chosen decision framework. D3A emphasizes validation, synchronization, and proportionality [2] in target selection and approval. Commanders, therefore, retain responsibility for alignment with intent and ensure that effects are legally and operationally appropriate [2].

Targeting decisions are institutionally structured to preserve human judgment and accountability. Legal review and compliance with the principles of the Law of Armed Conflict (LOAC) - distinction, proportionality, necessity, and humanity — are embedded throughout the process [3]. *D3A* is also iterative: outputs at one stage inform subsequent stages, allowing errors to propagate if not identified and corrected. Validation and cross-referencing are therefore not inefficiencies, but intentional safeguards. While targeting processes are an excellent use case for AI integration, such AI must operate within these doctrinal guardrails rather than bypass them.

Despite rapid technical progress and research on human-AI teaming in high-risk decision-making processes, the workflow-level operationalization of AI support for doctrinal targeting remains underdeveloped. Hybrid symbolic-semantic systems are the current state of the art for these systems and support relational reasoning and semantic retrieval [4]. Yet, few efforts embed AI assistance directly within doctrinal targeting stages or instrument those interactions for rigorous empirical evaluation.

This paper addresses this gap by developing a workflow-embedded cognitive architecture explicitly aligned with *D3A* stages as articulated in FM 3-60. Figure 1 illustrates the proposed workflow. It is not presented as an autonomous targeting engine, but rather as bounded analytic decision support operating within a defined doctrinal structure. Relational knowledge representation, semantic retrieval mechanisms, and role-configured agents are integrated through constrained interfaces that preserve command authority and support trace capture. Architecturally, the system demonstrates how AI components can be embedded within doctrinal decision stages without displacing command responsibility. It establishes a framework for empirically evaluating how such embedding influences task efficiency, alignment with commander guidance, cognitive workload, and trust calibration under simulated operational conditions. By situating AI integration at the intersection of doctrine, governance, and cognitive workflow design, this research reframes automation toward accountable augmentation in high-consequence environments.

II. Related Work

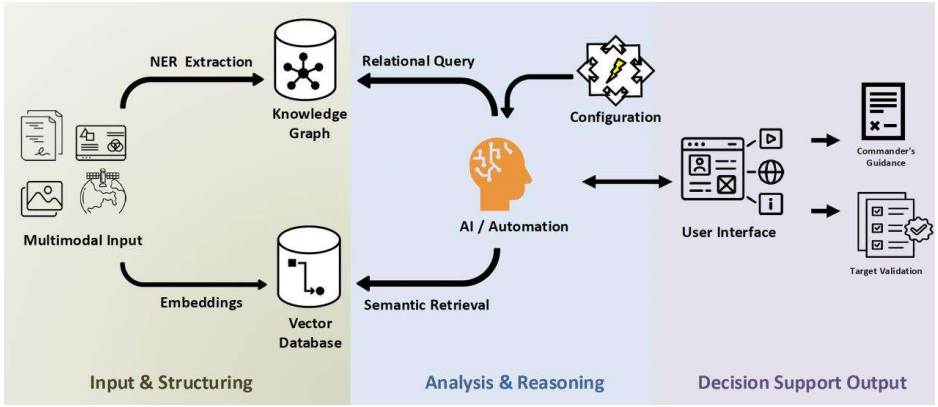
Research on artificial intelligence in military decision environments spans governance guidance for AI-enabled systems, studies of human–AI interaction in high-stakes workflows, and technical architectures for information retrieval and reasoning. Prior work highlights the importance of maintaining human involvement and traceability when AI systems assist operational decision processes. However, much of this research examines policy guidance or technical capabilities separately, leaving

Year Two

Breakdown of sub-efforts

Parent Project

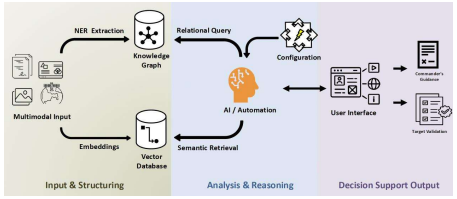
Exploring AI and Targeting: Explore how AI can support targeting workflow related tasks without displacing human judgement, by embedding AI within structured checkpoints that enable review, verification, and traceability.



Extension and Breakdown of main project to push the state-of-the-art on each sub-efforts

Extension and breakdown: Five sub-efforts

Exploring AI and Targeting: Explore how AI can support targeting workflow related tasks without displacing human judgement, by embedding AI within structured checkpoints that enable review, verification, and traceability.



1. **Preserving Critical Details from Mixed-Format Inputs:** Loss-aware multimodal data wrangling in AI-enabled targeting to ensure orders, imagery, ISR, tables, and overlays are ingested with minimal information loss for downstream decision support. Hybrid multimodal extraction with stage-wise information loss attribution and comparative evaluation across retrieval modes (RAG / RAG + KG / KG-only)

2. **Catching Staleness in the Operational Picture:** Auditable cross-database validation in AI-enabled targeting to ensure command decisions are based on current intelligence, analysis, and priorities with a defensible audit trail. Knowledge graph as temporal validity layer over the vector database for semantic extraction; multi-mode routing to ensure speed and accuracy while conserving audit provenance.

3. **Target Validation in AI-enabled Targeting:** Target Validation in AI-enabled Targeting to ensure proposed targets satisfy commander's intent, institutional knowledge, intelligence, and doctrinal and legal criteria before action. AI-assisted cross-checking surfaces source-attributed evidence per criterion and produces PASS/FAIL artifacts for legal and command review.

4. **Making AI Uncertainty Visible in Targeting:** Visualizing AI Uncertainty in Targeting to help decision-makers see when the AI is unsure and trace where information came from. Workflow-centric, transformation-aware indicators surface uncertainty signals from AI and Automation to visually convey the information needed at each decision checkpoint while preserving provenance across encodings

5. **Exploring Governance in AI-enabled Targeting Workflows:** Explores AI governance, Law of War, and Meaningful Human Control principles for layered control modes at targeting decision checkpoints; tested through instrumented user studies with distinct research questions across all sub-efforts to inform refinement

Preserving Critical Details from Mixed-Format Inputs:
Loss-aware data wrangling

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Problem Statement

- **State of the Art:** Multimodal RAG pipelines handle text, tables, and figures but stage-wise information loss is poorly understood. Benchmarks assess retrieval performance, not ingestion fidelity.
- **Gaps:** No method ensures multimodal military inputs (orders, ISR, imagery, tables) are ingested with minimal loss. What is lost, where, and of what type remains undetectable in current systems and how to overcome the loss of information.
- **Possible Solutions:** Hybrid ingestion combining text and visual extraction paths with stage-wise loss instrumentation and a test capability to surface loss awareness across element types and retrieval modes (RAG-only, RAG with KG validation, KG-only).

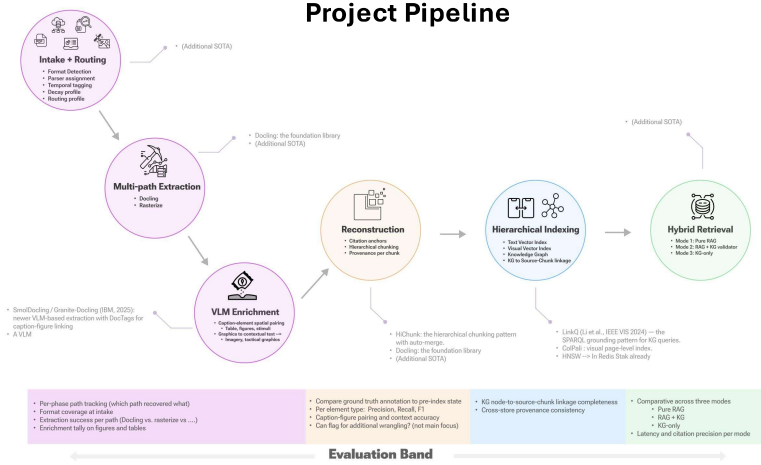
Project Deliverables

- **Stakeholder(s): DAMO-SO, ARCYBER, DA G2**
- **Deliverable Products:**
  - Publications and technical reports based on findings.
  - Hybrid extraction pipeline
  - Military document benchmark and comparative result
- **Expected Outcomes:**
  - Exploration results of loss awareness methodology
  - Comparative findings across retrieval modes informing downstream system design

Project Description

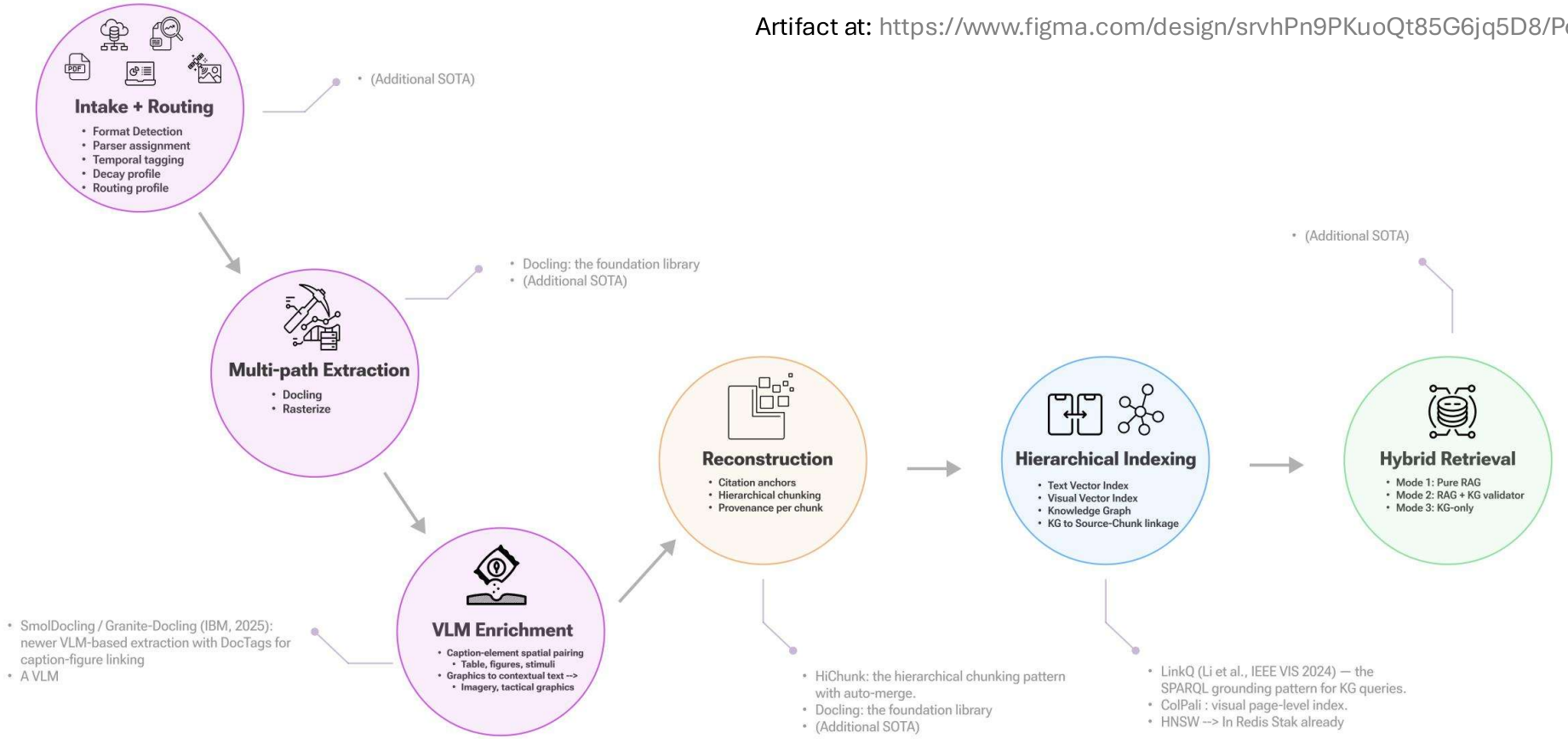
- **Goal**: Explore hybrid multimodal ingestion to begin ensuring inputs are processed with minimal loss while building awareness of what is lost and where.
- **Methods**:
  - VLM-enriched extraction with paired captions and section paths
  - Stage-wise loss measurement stratified by element type
  - Comparative evaluation across RAG-only, RAG with KG validator, and KG-only modes
- **Implications**: Works toward ingestion capability for high-stakes multimodal corpora. Informs which approach minimizes loss for which element type.

Project Pipeline



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Artifact at: https://www.figma.com/design/srvhPn9PKuoQt85G6jq5D8/Poster



- Per-phase path tracking (which path recovered what)
- Format coverage at intake
- Extraction success per path (Docling vs. rasterize vs ....)
- Enrichment tally on figures and tables

- Compare ground truth annotation to pre-index state
- Per element type: Precision, Recall, F1
- Caption-figure pairing and context accuracy
- Can flag for additional wrangling? (not main focus)

- KG node-to-source-chunk linkage completeness
- Cross-store provenance consistency

- Comparative across three modes
  - Pure RAG
  - RAG + KG
  - KG-only
- Latency and citation precision per mode

Evaluation Band

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Catching Staleness in the Operational Picture: *Cross-database validation*

Problem Statement

- **State of the Art:** Hybrid retrieval systems combining vector search and knowledge graphs are well explored. Systems such as LinkQ, RouteRAG, HybGRAG, and SymRAG use knowledge graphs as alternative retrieval sources alongside vector databases.
- **Gaps:** Vector databases cannot represent supersession or temporal validity. As new intel arrives, stale chunks are retrieved alongside current ones.
- **Possible Solutions:** Three-mode architecture with intent-based routing: pure RAG for open-ended queries, RAG with KG staleness validator for entity-dependent queries, and direct KG queries for structured relational queries. Knowledge graph nodes carry validity fields and supersession chains preserving full audit history.

Project Deliverables

- **Stakeholder(s): DAMO-SO, ARCYBER, DA G2**
- **Deliverable Products:**
  - Publications and technical reports based on findings.
  - Cross-database validator service.
- **Expected Outcomes:**
  - Initial evidence on staleness detection across retrieval modes

Project Description

- **Goal**: Explore cross-database validation to begin detecting temporal drift across hybrid AI knowledge stores while maintaining a defensible audit trail.
- **Methods**:
  - Schema-induced knowledge graph with temporal validity fields and supersession chains
  - Validator service cross-checking retrieved chunks against current knowledge graph entity state
  - Comparative evaluation of retrieval accuracy and staleness detection across three modes
- **Implications**: Informs the design of knowledge graph as a temporal validity layer.

Example Use Case for Design and Usability Studies

- **Stale Intel Detection:** Target validation requires up-to-date target coordinates at the moment of validation. Stale target locations can become confused with more recent ones as they are added to the KG. Recent intel must be prioritized over older intel in these processes.

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Target Validation in AI-enabled Targeting:
Verified targets, traceable evidence

Problem Statement

- **State of the Art:** AI and automation are increasingly used in military decision workflows; agentic systems chain reasoning, evaluate evidence, and route between knowledge stores. JTC defines multi-criterion target validation, providing opportunity to apply agentic AI with traceable evidence.
- **Gaps:** Application of agentic AI for criterion-based target validation with source-attributed evidence inside doctrinal workflows is not yet well studied.
- **Possible Solutions:** Building on multimodal data wrangling and cross-database validated knowledge, explore agentic cross-checking of proposed targets against multi-source criteria toward PASS/FAIL artifacts with traceable evidence for legal and command review.

Project Deliverables

- **Stakeholder(s): DAMO-SO, ARCYBER, DA G2**
- **Deliverable Products:**
  - Publications and technical reports based on findings.
  - Target validation workflow prototype
- **Expected Outcomes:**
  - Initial validation workflow producing defensible artifacts
  - Criterion-based methodology for doctrinal target validation

Project Description

- **Goal**: Explore how agentic AI can support multi-criterion target validation using multimodal and cross-validated knowledge, producing defensible evidence for legal and command review.
- **Methods**:
  - Define criteria schema across commander's intent, intelligence, institutional knowledge, and doctrinal/legal requirements
  - Build agentic cross-checking pipeline over ingested and cross-database validated knowledge to account for temporal decay
  - Surface source-attributed evidence per criterion
  - Produce PASS/FAIL artifacts with reasoning trail
- **Implications**: Informs how agentic AI supports multi-criterion target validation without replacing analyst judgment.

Example Use Case for Design and Usability Studies

- **Pre-Nomination Target Validation:** Analyst reviews proposed target as AI surfaces evidence per criterion. Agents are assigned functional roles (e.g., Intel targeteer, legal oversight, etc) and produce targets that meet their role-based selection criterion. A master agent orchestrator combines these nominations into a valid target nomination for human review.

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Prototype focused on design study

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Making AI Uncertainty Visible in Targeting:
Uncertainty and evidence made visible

Problem Statement

- **State of the Art:** Uncertainty visualization research shows visualized uncertainty improves trust calibration and reliance. Multiple coordinated views and provenance visualization provide established design vocabulary.
- **Gaps:** Existing AI uncertainty visualizations are model-centric. None are workflow-centric or tied to decision checkpoints. Chart transformations across roles and what provenance to preserve across encodings remain unexplored for AI-assisted high-stakes decisions.
- **Possible Solutions:** Build workflow-centric, transformation-aware indicators that surface uncertainty signals from AI and Automation. Design alternative encodings tuned to analyst and commander mental models. Pre-register which provenance fields persist across transformations for legal review.

Project Deliverables

- **Stakeholder(s): DAMO-SO, ARCYBER, DA G2**
- **Deliverable Products:**
  - Publications and technical reports based on findings.
  - Indicator and chart-transformation design library
  - Design guidelines for uncertainty communication
- **Expected Outcomes:**
  - Empirically validated uncertainty signals across generation, retrieval, semantic, model, and workflow layers, mapped to factual errors and hallucinations
  - VIS techniques validated for trust calibration & detection accuracy

Project Description

- **Goal**: Visually convey the information operators and decision-makers need at each decision checkpoint while preserving provenance across encodings.
- **Methods**:
  - Identify uncertainty signals from upstream components (extraction loss, cross-database validator output, governance flags).
  - Design indicators aligned with operator and CDR's mental models.
  - Run user and design studies on trust calibration, detection accuracy, decision speed, and cognitive load.
- **Implications**: Shifts uncertainty communication from a model-output property to a workflow-aware decision aid. Identifies which provenance is load-bearing for legal review and which can be summarized without loss of accountability.

Example Use Case for Design and Usability Studies

- **SOC Operations Floor:**
  - Live agentic updates to SOC big-screen intelligence estimates render quietly for the floor while selectively alerting the responsible analyst, whose workstation reveals layered uncertainty for audit and verification.
- **Analyst Reporting Pipeline :** During report drafting, claim-level uncertainty surfaces weak or low-agreement claims for analyst review and persists into the published product as an audit trail for downstream consumers .

Prototypes focused on design study

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Exploring Governance in AI-enabled Targeting:
Layered control, defensible decisions

Problem Statement

- **State of the Art:** Military AI governance draws on domestic and international policy and principles for responsible AI and weapons development. Automated weapons policy uses the meaningful human control (MHC) principles and human-automation control theory, but they cannot be easily applied to AI-enabled weapons.
- **Gaps:** Undefined terms like MHC and effective oversight are the current norm in AI governance, which cannot be translated into checkpoint-level instrumentation.
- **Possible Solutions:** Map governance principles to decision checkpoints within doctrinal targeting workflows; staged levels-of-automation framing defines control modes and required artifacts.

Project Deliverables

- **Stakeholder(s): DAMO-SO, ARCYBER, DA G2**
- **Deliverable Products:**
  - Publications and technical reports based on findings.
  - Initial layered control framework
  - Empirical findings on leader comfort across checkpoints and echelons
- **Expected Outcomes:**
  - Preliminary mapping of gov. principles to operational checkpoints
  - Empirical insight into human-AI interaction at decision checkpoints
  - Identification of feasibility limits, friction points, and oversight needs

Project Description

- **Goal**: Explore AI governance, Law of War, and MHC principles to begin translating them into stage-specific layered control aligned with doctrinal targeting; test through focused user studies across all sub-efforts.
- **Methods**:
  - Understand MHC, Law of War, DoD RAI, and NIST RMF requirements
  - Design layered visualizations: token encoding, claim-level confidence bands, provenance-linked indicators, and uncertainty-triggered alerts that route to the responsible analyst
  - Run user and design studies on trust calibration, detection accuracy, decision speed, and cognitive load
- **Implications**: Explores "human (x) the loop" with stage-specific layered control. Generates evidence feeding back into all sub-efforts

Example Use Case for Design and Usability Studies

- **Layered Control Across Targeting Cycle:** Leaders at different echelons interact with the same proposed target as AI shifts between advisory, co-decision, and authoritative control modes at each checkpoint. Surfaces where leaders accept AI involvement and where they retain authority.

- **Edge Case Escalation and Legal Review:** AI detects ambiguity (civilian presence, conflicting intel) and triggers escalation with required logged artifacts.

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